Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
2	(a)		Correct expression for centripetal force used (accept acceleration if equated to gravitational field strength) (1) accept with ω^2 Equating to gravitational expression i.e. $\frac{mv^2}{r} = \frac{GMm}{r^2}$ **or** $\frac{v^2}{r} = \frac{Gm}{r^2}$ (1) Enough algebra shown. Minimum is this: $\frac{mv^2}{r} = \frac{GMm}{r^2} \quad v^2 = \frac{GM}{r}$ (1) **Alternative 1:** $m\omega^2 r$ **or** $\omega^2 r$ quoted (1) $m\omega^2 r = \frac{GMm}{r^2}$ **and** $v = \omega r$ **or** acceleration equivalent (1) Minimum algebra leading to answer (1) **Alternative 2:** Using M_1 or M_2 negligible in $T = 2\pi\sqrt{\frac{d^3}{G(M_1+M_2)}}$ (1) e.g. $2\pi\sqrt{\frac{d^3}{G(M_1+M_2)}} \approx 2\pi\sqrt{\frac{d^3}{GM}}$ Quoting $v = \omega r$ and $\omega = \frac{2\pi}{T}$ (1) or equivalent All algebra ok leading to the answer (1)	3			3	2	
	(b)		Measured velocities are too large (1) Accept velocity increasing on graph Link to the equation e.g. velocity increases with mass (1) Dark matter accounts for increased mass **or** the mass must be bigger (1)	3			3		

Question			Marking details	AO1	AO2	AO3	Total	Maths	Prac
				Marks available					
	(c)		Rearrangement for M i.e. $M = \frac{v^2 r}{G}$ (1) Conversion of distance i.e. 20 000 × 9.46 × 10^{15} = (1.892 × 10^{20}) (1) Answer = 7.09 × 10^{39} k[g] (1)		3		3	3	
	(d)		Speed estimated as 110 - 130 [km s^{-1}] (1) Use of the Doppler equation **ecf** (1) Answer = 7.7 × 10^{-5} to 9.1 × 10^{-5} [m] (1)	1	1 1		3	2	
	(e)		Use of Hubble equation $v = H_0 d$ (1) Answer = 5.0 × 10^{22} to 5.9 × 10^{22} [m] (**ecf** on v) (1)	1	1		2	2	
			Question 2 total	**8**	**6**	**0**	**14**	**9**	**0**